NEURO 208, School of Medicine, University of Crete

Lecture 3: Synapses

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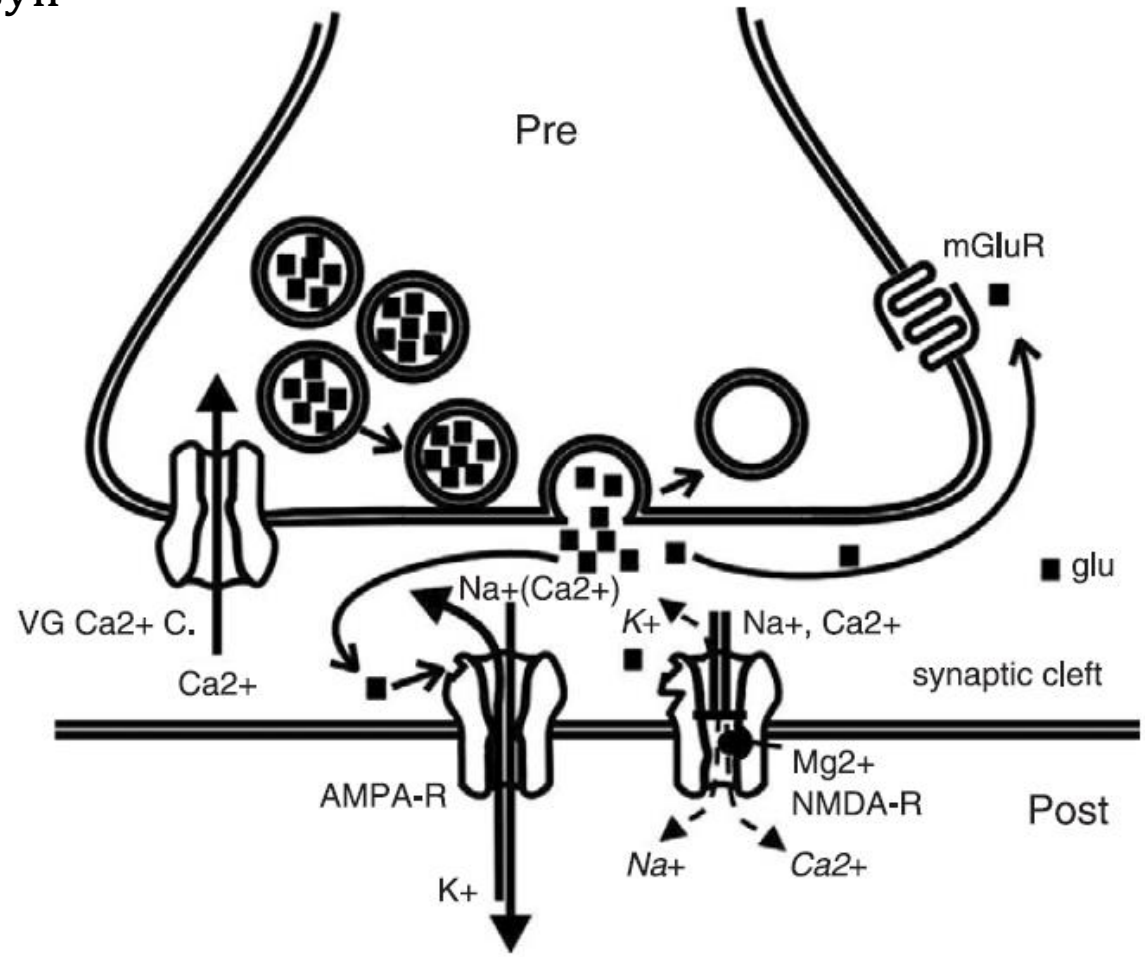
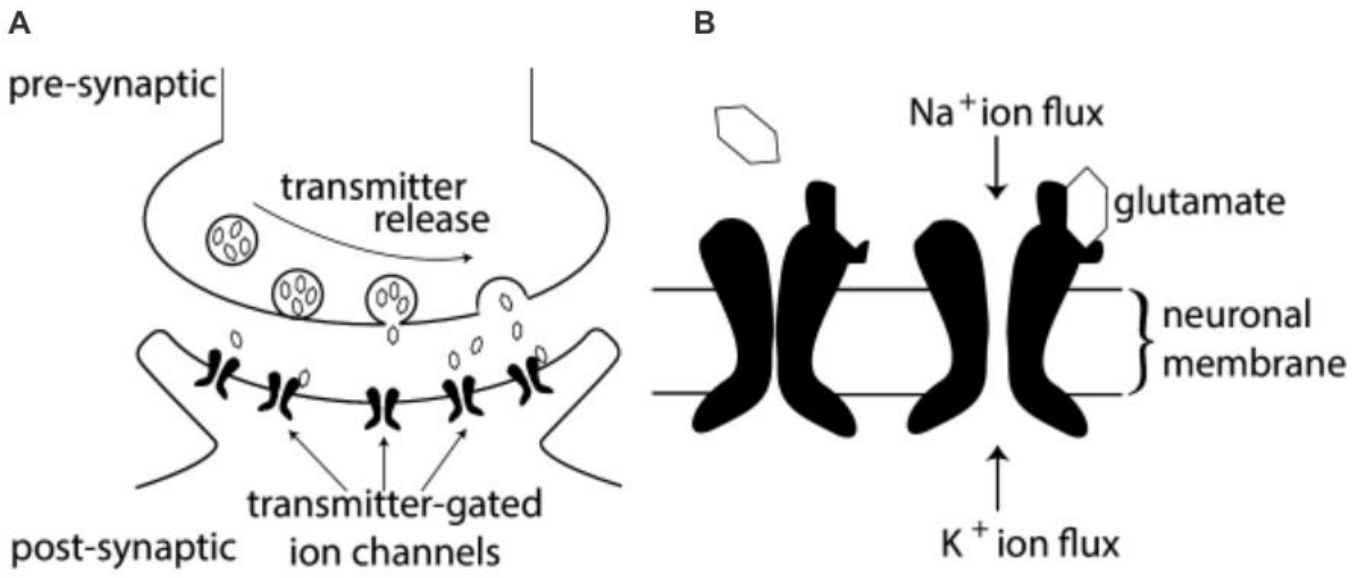
IMBB-FORTH

Modelling Synapses

What about I_{syn} ?



Synaptic transmission



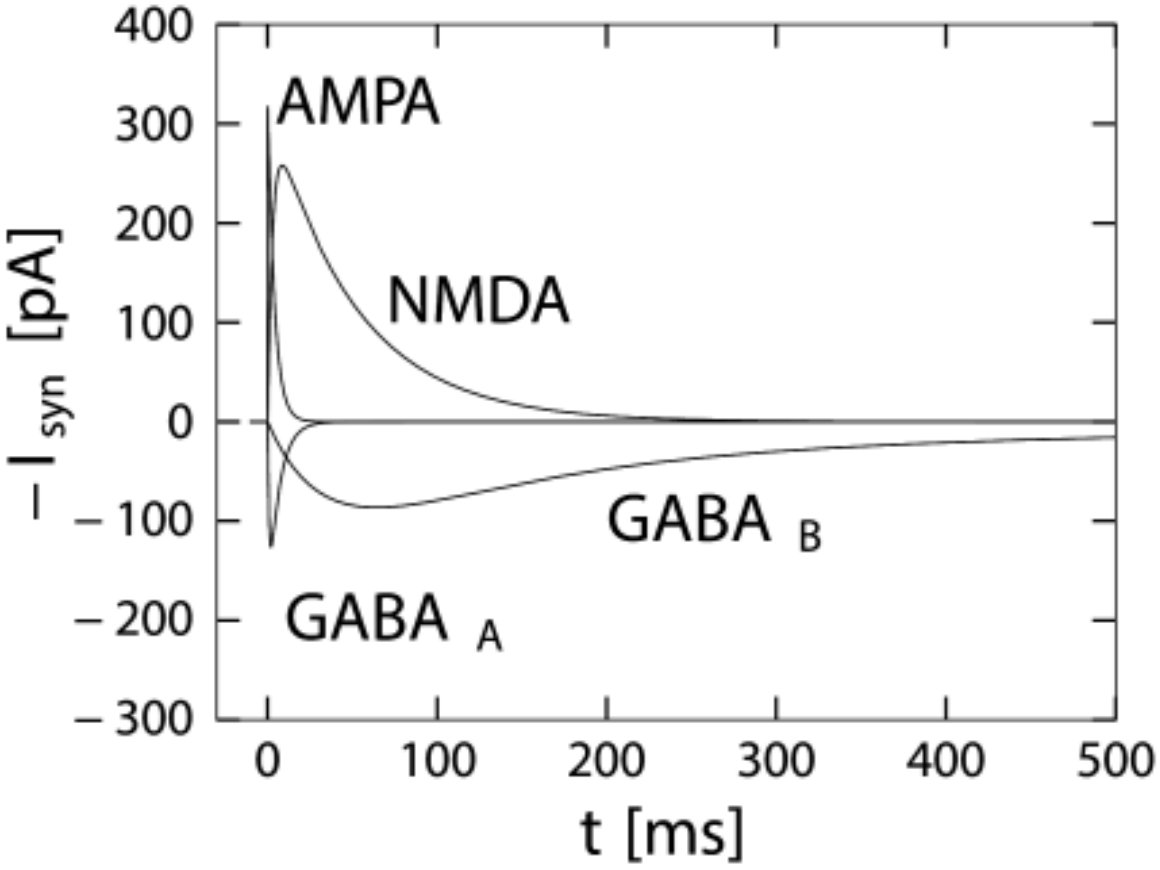
Adopted from [Gerstner, Kistler, Naud, Paninski, Neuronal Dynamics]

- Upon arrival of a presynaptic spike, neurotransmitter spills into the synaptic cleft and is captured by postsynaptic receptors.
- When glutamate is bound to the receptor, sodium and potassium ions can flow through the membrane.

Dynamics of post-synaptic current

- Excitatory (glutamate)
  - AMPA (fast)
  - NMDA (slow)

- Inhibitory (GABA)
  - $GABA_A$ (fast)
  - $GABA_B$ (slow)



Adopted from Gerstner, Kistler, Naud, Paninski, Neuronal Dynamics

Conductance-Based and Current-Based Synapses

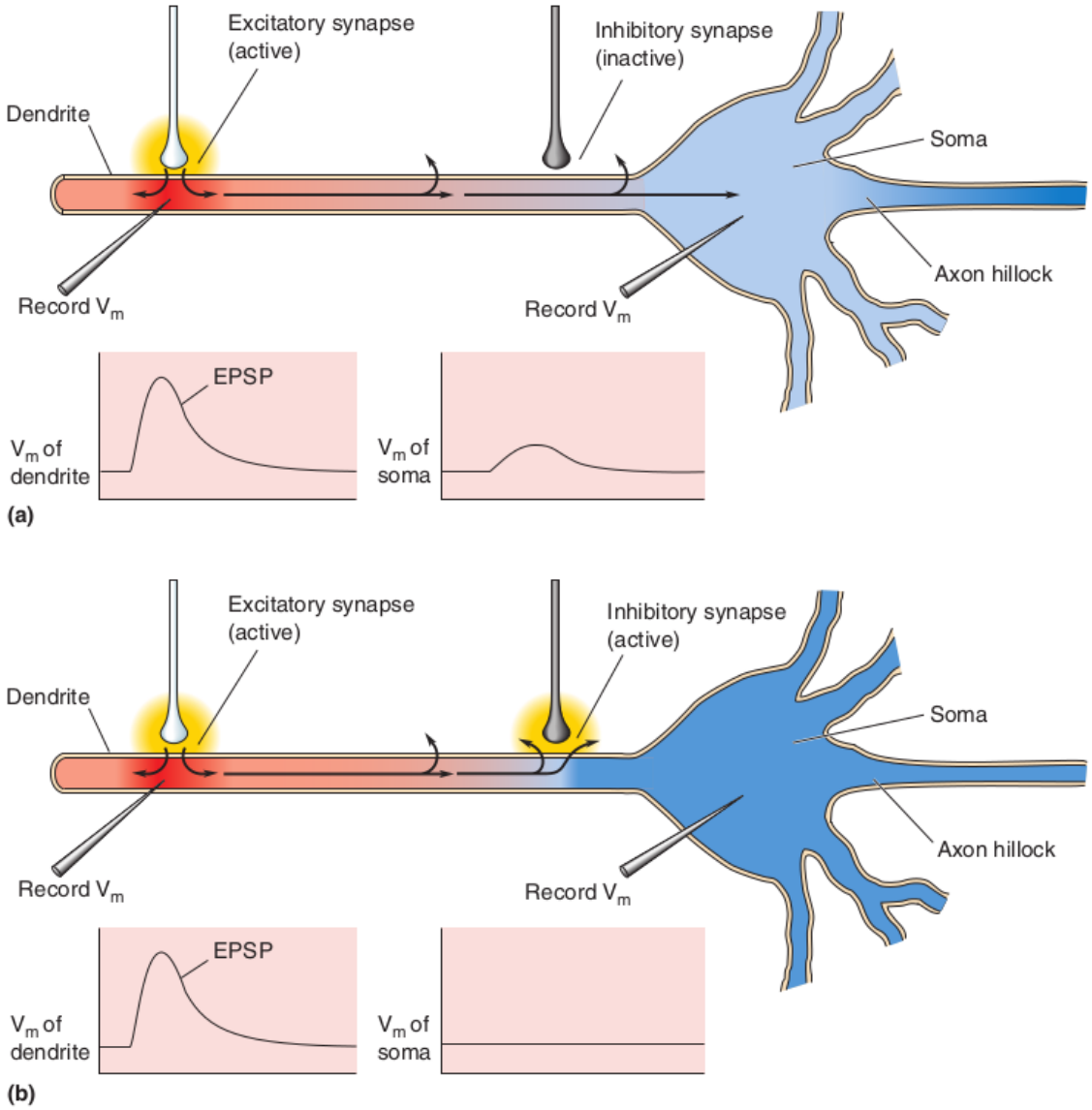
$$I_{syn} = \mathrm{g}_{syn}(t)\big(V - E_{syn}\big)$$

$$C\frac{dV}{dt} = -\bar{\mathrm{g}}_L(V - E_L) - \mathrm{g}_E(t)(V - E_E) - \mathrm{g}_I(t)(V - E_I)$$

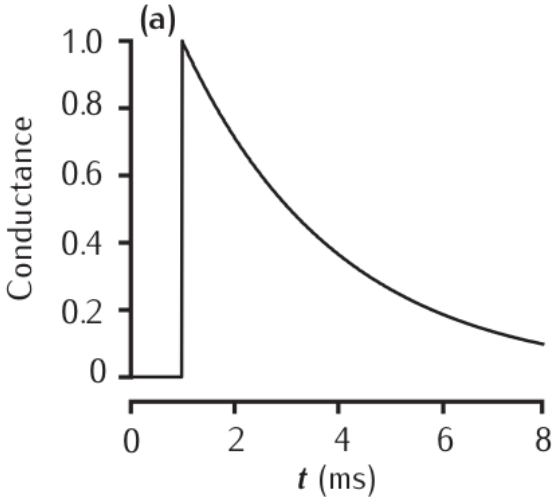
- What happens in steady state?

- Current based synapses: Synapses as sources of current and not a conductance, i.e., without the dependence on the membrane potential V (e.g., $V = V_{rest}$)

Shunting Inhibition

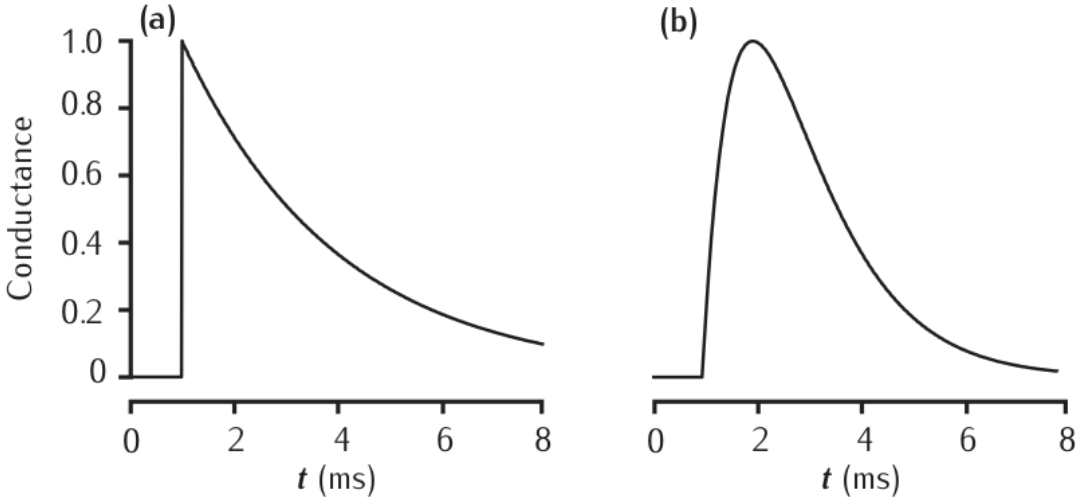


Simple conductance waveforms



(a) $\mathrm{g_{syn}}(t) = \bar{\mathrm{g}}_{\mathrm{syn}} \exp\left(-\frac{t - t_{\mathrm{spike}}}{\tau}\right) \Theta\left(t - t_{spike}\right)$

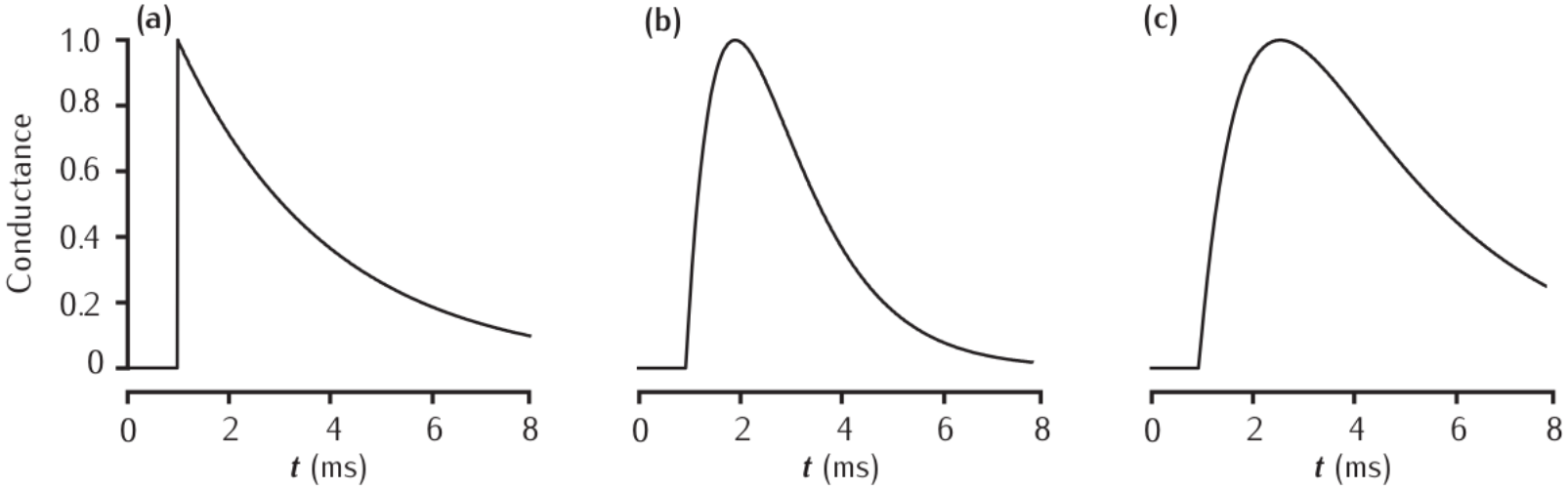
Simple conductance waveforms



(a) $g_{\mathrm{syn}}(t) = \bar{g}_{\mathrm{syn}} \exp\left(-\frac{t - t_{\mathrm{spike}}}{\tau}\right) \Theta(t - t_{spike})$

(b) $g_{\mathrm{syn}}(t) = \bar{g}_{\mathrm{syn}} \frac{t - t_{\mathrm{spike}}}{\tau} \exp\left(\left(1 - \frac{t - t_{\mathrm{spike}}}{\tau}\right)\right) \Theta(t - t_{spike})$

Simple conductance waveforms



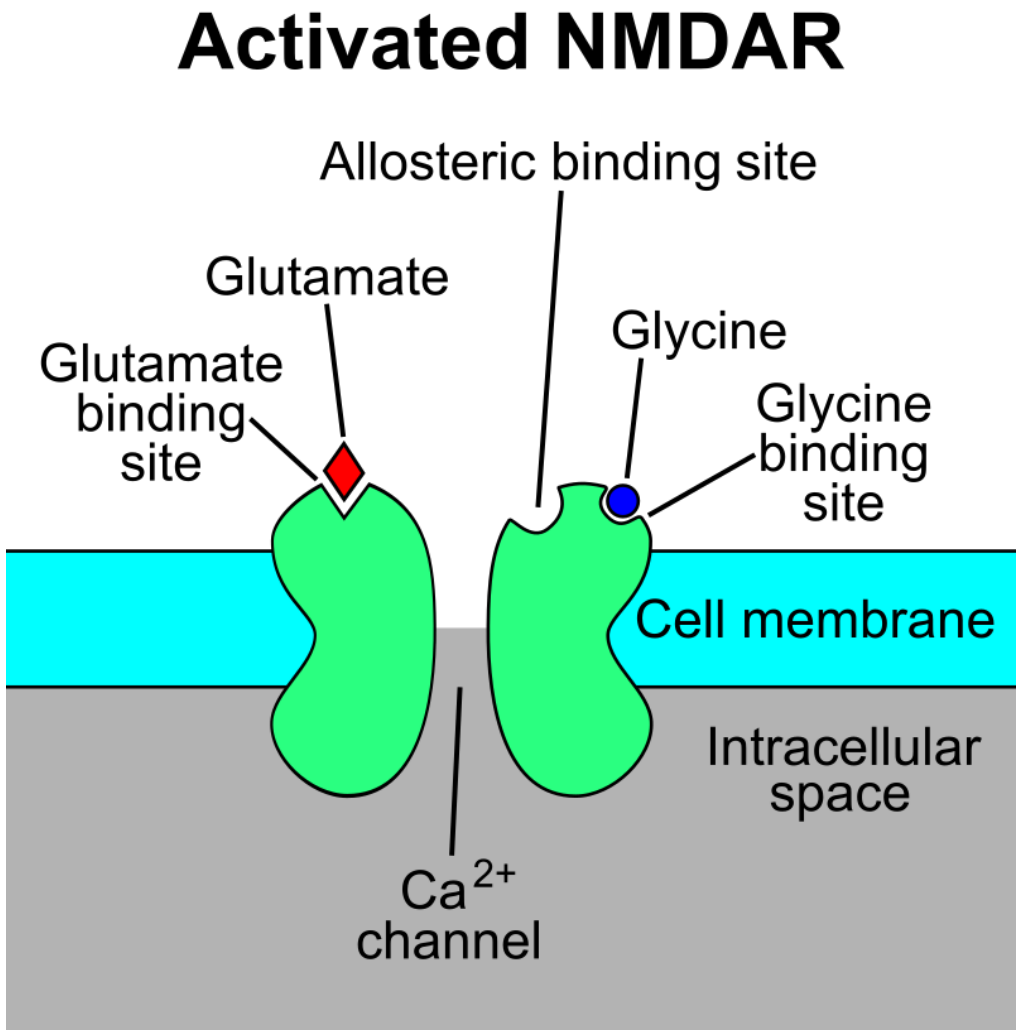
(a) $g_{\mathrm{syn}}(t) = \bar{g}_{\mathrm{syn}} \exp\left(-\frac{t - t_{\mathrm{spike}}}{\tau}\right) \Theta(t - t_{spike})$

(b) $g_{\mathrm{syn}}(t) = \bar{g}_{\mathrm{syn}} \frac{t - t_{\mathrm{spike}}}{\tau} \exp\left(\left(1 - \frac{t - t_{\mathrm{spike}}}{\tau}\right)\right) \Theta(t - t_{spike})$

(c) $g_{\mathrm{syn}}(t) = \bar{g}_{\mathrm{syn}} \frac{\tau_d \tau_r}{\tau_d - \tau_r} \left(\exp\left(-\frac{t - t_{\mathrm{spike}}}{\tau_d}\right) - \exp\left(-\frac{t - t_{\mathrm{spike}}}{\tau_r}\right)\right) \Theta(t - t_{spike})$

Voltage-Dependent Synapse

- Conduct Ca^{2+}
- Critical role for long-term modification of synaptic strength
- Act as coincidence detectors; pre- and post- activity

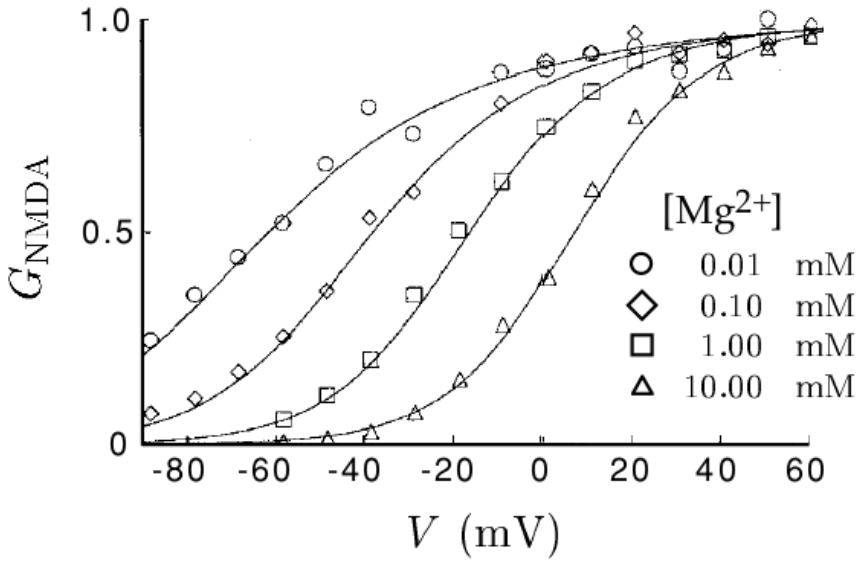


Adopted from https://en.wikipedia.org/wiki/NMDA_receptor

NMDA model

$$g_{\mathrm{NMDA}}(t) = g_{\mathrm{syn}}(t) f(V)$$

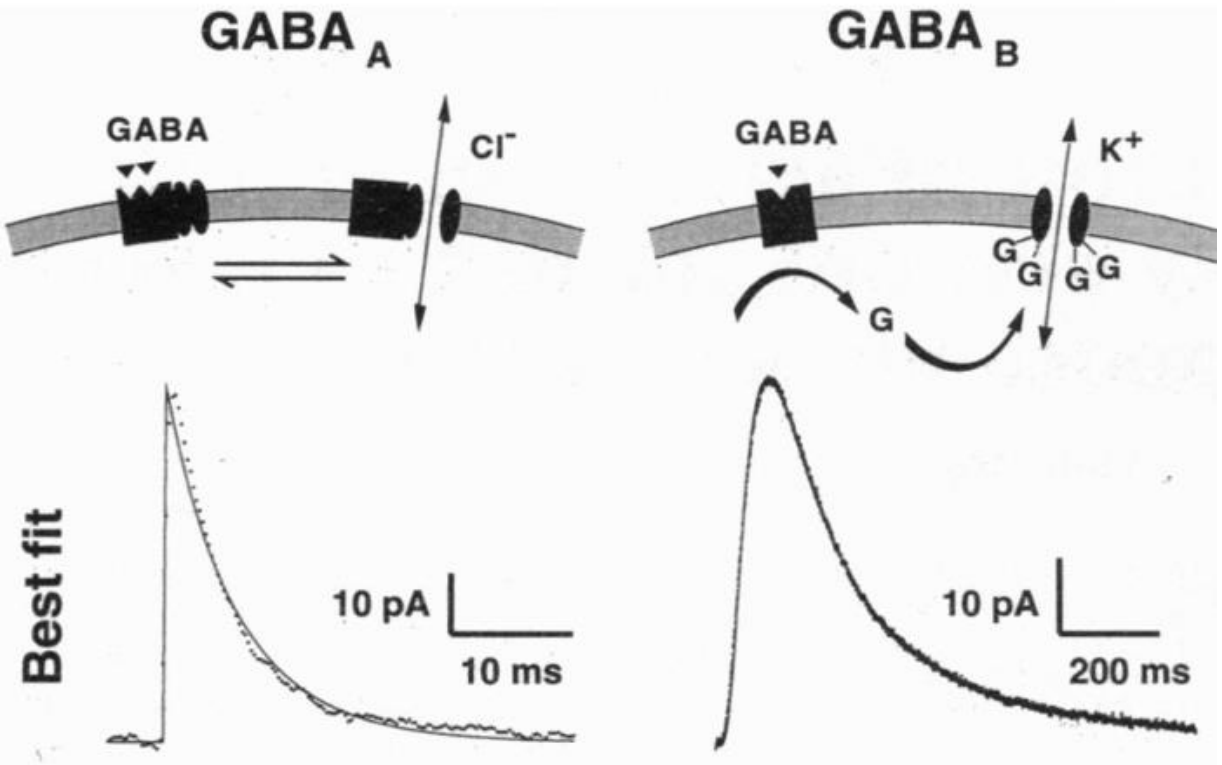
$$f(V) = \left(1 + \frac{[\mathrm{Mg}^{2+}]}{3.57\mathrm{mM}} \exp(-V/16.13\mathrm{mV})\right)^{-1}$$



Adopted from Jahr and Stevens, 1990

Second Messenger-Gated Ion Channels

- Have more complex scheme of activation involving the activation of K^+ channels by G proteins.
- Synaptic activation results in a slow inhibitory postsynaptic potential which involves an increase in K^+ conductance.
- The reversal potential of $GABA_B$ is lower than this of $GABA_A$.



Adopted from Destexhe and Sejnowski, 1995

G-protein cascade

- Based on cooperativity in the activation of K^+ channels, which need several G-protein binding to be activated (Destexhe et al., 1994)

$$
\begin{aligned}
&R_0 + T \rightleftarrows R \rightleftarrows D \\
&R + G_0 \rightleftarrows RG \rightarrow R + G \\
&G \rightarrow G_0 \\
&\mathrm{C} + nG \rightleftarrows \mathrm{O}
\end{aligned}
$$

- Transmitter T binds to the receptor R_0 leading to its activation form R, and desensitized form D.
- The G protein is transformed from an inactive form G_0, to an activated form G, catalyzed by R.
- G binds to open the K^+ channel with n independent binding sites

$GABA_B$ model (1)

- Assume quasi-stationary in G-related equations (2 and 3) and consider G_0 in excess (Destexhe and Sejnowski, 1995)

$$\frac{d[R]}{dt} = K_1 \cdot [T] \cdot (1 - [R] - [D]) - K_2 \cdot [R] + d_2 \cdot [D]$$

$$\frac{d[D]}{dt} = d_1 \cdot [R] - d_2[D]$$

$$\frac{d[G]}{dt} = K_3 \cdot [R] - K_4 \cdot [G]$$

At t_{pre}: $[T] \rightarrow [T] + [T]_{\max}$

At $t_{pre^{last}} + t_{dur}$: $[T] \rightarrow 0$

$$I_{\mathrm{GABA_B}} = \bar{g}_{\mathrm{GABA_B}} \cdot \frac{[G]^n}{[G]^n + K_D} \cdot (V - E_{\mathrm{K}})$$

- $[R]$: fraction of activated receptors, unitless
- $[D]$: fraction of desensitized receptor
- $[G]$: concentration of activated G-protein (mM)
- K_D: dissociation constant of the binding of G on the K^+ channels
- [T]: transmitter concentration (mM)
- n: number of binding sites of G-protein on K^+, $n = 4$.
- K_j, d_j: kinetic rate constants
  - K_1: forward binding rate to receptor ($\mathrm{ms}^{-1}\ \mathrm{mM}^{-1}$)
  - K_2: backward (unbinding) rate of receptor (ms^{-1})
  - K_3: rate of G-protein production (ms^{-1})
  - K_4: rate of G-protein decay (ms^{-1})
  - d_1: rate of desensitization (ms^{-1})
  - d_2: rate of re-sensitization (ms^{-1})

$GABA_B$ model (2)

- Assume quasi-stationary in G-related equations (2 and 3) and consider G_0 in excess (Destexhe et al., 1996) – Assume also no desensitization

$$\frac{d[R]}{dt} = K_1 \cdot [T] \cdot (1 - [R]) - K_2 \cdot [R]$$

$$\frac{d[G]}{dt} = K_3 \cdot [R] - K_4 \cdot [G]$$

At t_{pre}: $[T] \rightarrow [T] + [T]_{\max}$

At $t_{pre^{last}} + t_{dur}$: $[T] \rightarrow 0$

$$I_{\mathrm{GABA_B}} = \bar{g}_{\mathrm{GABA_B}} \cdot \frac{[G]^n}{[G]^n + K_D} \cdot (V - E_{\mathrm{K}})$$

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  - K_3: rate of G-protein production (ms^{-1})
  - K_4: rate of G-protein decay (ms^{-1})

$GABA_B$ model (3) - GABA

- Diffusion of GABA in the extracellular space (Thomson and Destexhe, 1999)
- GABA uptake was simulated according to the following scheme

$$T + U \rightleftarrows B$$
$$B \rightarrow U + \mathrm{GABA_i}$$

- The first step describes the **reversible** binding of GABA (T) to the unbound transporter (U), leading to bound transporter (B).
- The second step represents the **irreversible** translocation of GABA into the intracellular space ($\mathrm{GABA_i}$)

$$\frac{d[T]}{dt} = -k_1 \cdot [T] \cdot (B_m - [B]) + k_{-1}[B] - \frac{[T]}{\tau_D}$$

$$\frac{d[B]}{dt} = k_1 \cdot [T] \cdot (B_m - [B]) - (k_{-1} + k_2)[B]$$

- $[T]$: concentration of GABA
- $[B]$: concentration of bound transporter
- k_1 and k_{-1}: forward ($\mathrm{ms^{-1}mM^{-1}}$) and backward ($\mathrm{ms^{-1}}$) binding rate of GABA to transporter
- k_2: rate constant for the translocation of GABA ($\mathrm{ms^{-1}}$)
- B_m: total extracellular concentration of available transporter (mM)
-

$GABA_B$ model (3) - $GABA_B$

- Activation of $GABA_B$ responses by GABA was simulated using the following kinetic equations

$$R_0 + T \rightleftarrows R \qquad (1)$$
$$R + G_0 \rightleftarrows RG \rightarrow R + G \qquad (2)$$
$$G \rightarrow G_0 \qquad (3)$$
$$K_0 + nG \rightleftarrows K \qquad (4)$$

- GABA (T) binds to the inactive form of the $GABA_B$ receptor (R_0), leading to the activated form of the receptor (R) – reaction 1
- R catalyzes the formation of active G-proteins (G) from their inactive form (G_0) – reaction 2
- Activated G-proteins (G) is hydrolyzed to its inactive form (G_0) – reaction 3
- n activated G-proteins bind to the closed form of the K^+ channel (K_0), leading to opening of K^+ channels

$$\frac{d[R]}{dt} = K_1 \cdot [T] \cdot (1 - [R]) - K_2 \cdot [R]$$

$$\frac{d[G]}{dt} = K_3 \cdot [R] - K_4 \cdot [G]$$

$$I_{\mathrm{GABA_B}} = \bar{g}_{\mathrm{GABA_B}} \cdot \frac{[G]^n}{[G]^n + K_D} \cdot (V - E_{\mathrm{K}})$$

Full model 3

$$\frac{d[T]}{dt} = -k_1 \cdot [T] \cdot (B_m - [B]) + k_{-1}[B] - \frac{[T]}{\tau_D}$$

$$\boldsymbol{at\ t_{pre}\ [T] = [T] + [T]_{\mathrm{max}}}$$

$$\frac{d[B]}{dt} = -k_1 \cdot [T] \cdot (B_m - [B]) + (k_{-1} + k_2)[B]$$

$$\frac{d[R]}{dt} = K_1 \cdot [T] \cdot (1 - [R]) - K_2 \cdot [R]$$

$$\frac{d[G]}{dt} = K_3 \cdot [R] - K_4 \cdot [G]$$

$$I_{\mathrm{GABA_B}} = \bar{g}_{\mathrm{GABA_B}} \cdot \frac{[G]^n}{[G]^n + K_D} \cdot (V - E_{\mathrm{K}})$$

- [T]: transmitter concentration (mM)
- [B]: bound GABA transporter (mM)
- [R]: fraction of activated receptor
- [G]: normalized concentration of activated G-protein
- k_1 and k_{-1}: forward ($\mathrm{ms^{-1}mM^{-1}}$) and backward ($\mathrm{ms^{-1}}$) binding rate of GABA to/of transporter
- k_2: rate constant for the translocation of GABA ($\mathrm{ms^{-1}}$)
- B_m: total extracellular concentration of available transporter (mM)
- K_D: dissociation constant of the binding of G on the K^+ channels
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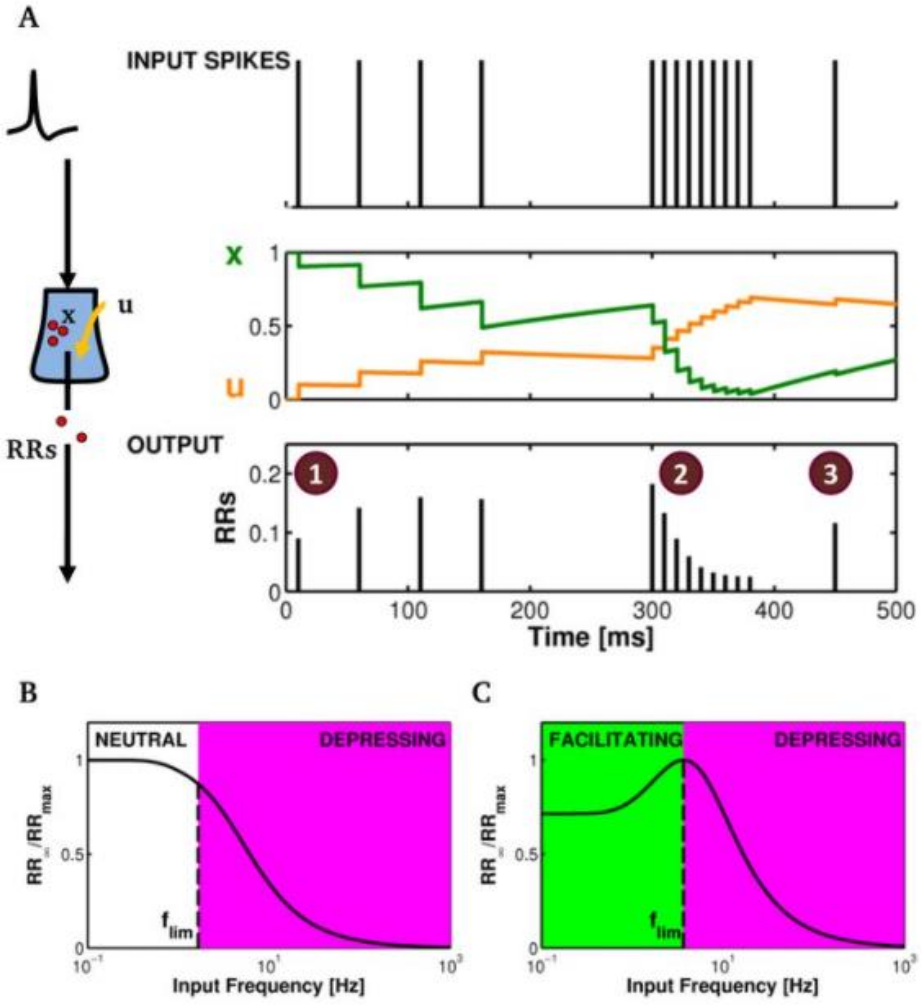
- K_3: rate of G-protein production ($\mathrm{ms^{-1}}$)
- K_4: rate of G-protein decay ($\mathrm{ms^{-1}}$)
- τ_D: decay of transmitter concentration (ms)

Dynamic synapses

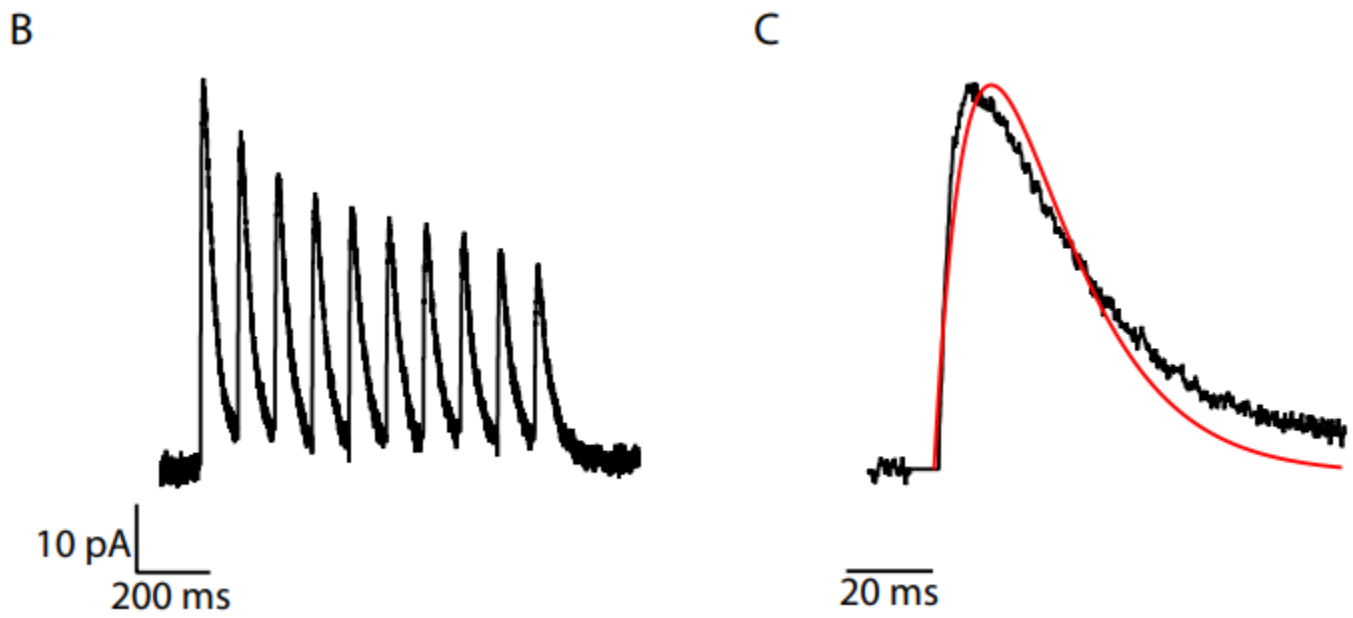
- **Short-Term Plasticity (STP)** is a phenomenon in which synaptic efficacy changes over time in a way that reflects the history of presynaptic activity.
- Two types of STP, with opposite effects on synaptic efficacy, have been experimentally observed.
- They are known as **Short-Term Depression (STD)** and **Short-Term Facilitation (STF)**.



Adopted from Pitta et al., 2011

Dynamic synapses

Short-Term Depression (STD): depletion of the pool of readily releasable vesicles can cause a decrease of the synaptic response to successive stimuli.



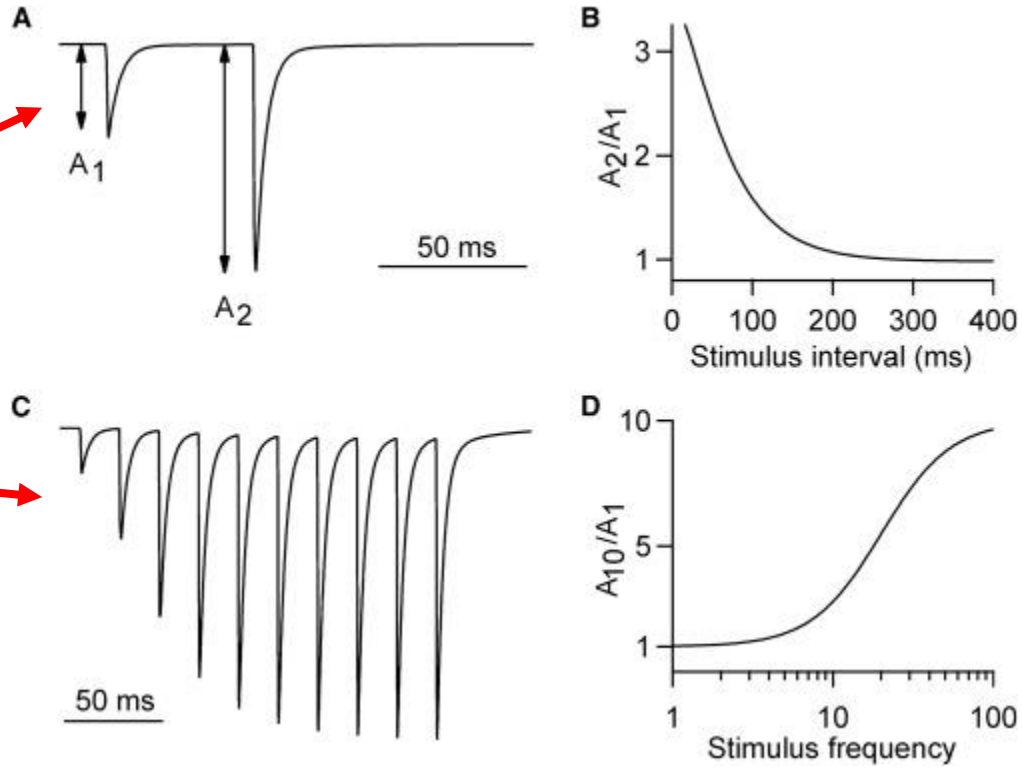
Adopted from Lavian and Korngreen, 2019

Dynamic synapses

Short-Term Facilitation (STF): Residual elevations of the presynaptic calcium concentration that are due to a previous action potential can increase the probability of subsequent transmitter release.

Pair pulses at 50 ms interval

50 Hz train of ten stimuli



Adopted from Jackman and Regehr, 2017

Phenomenological model

- The mathematical (http://www.scholarpedia.org/article/Short-term_synaptic_plasticity) of STP is based on the concept of a limited pool of synaptic resources available for transmission (x)
  - for example, the overall amount of synaptic vesicles at the presynaptic terminals.
- The amount of presynaptic resource changes in a dynamic fashion depending on the recent history of spikes.

Phenomenological model

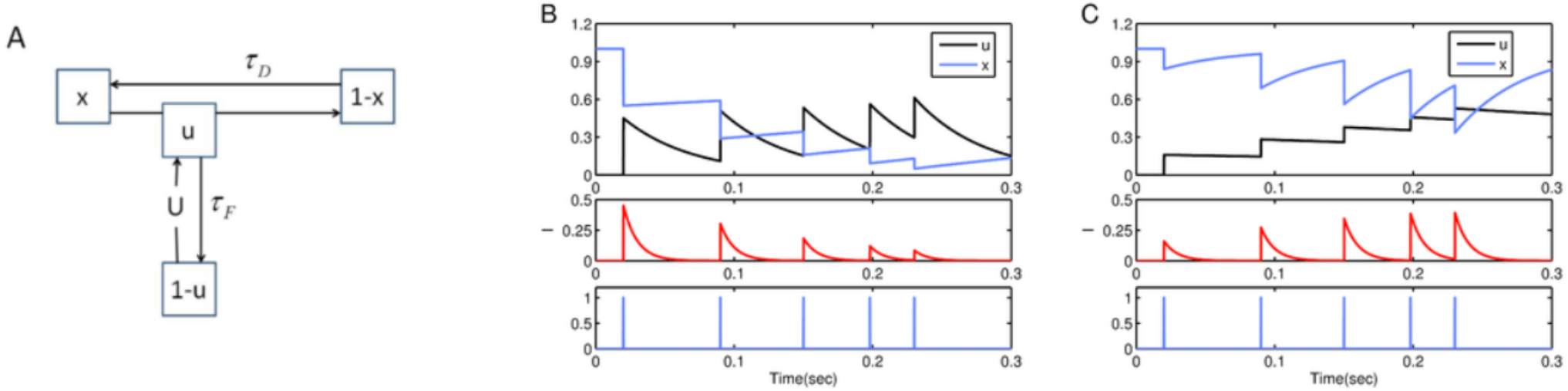
$$\frac{du}{dt} = -\frac{u}{\tau_f} + U_0(1 - u^-)\delta\left(t - t_{sp}\right)$$

$$\frac{dx}{dt} = \frac{1 - x}{\tau_d} - u^+ x^- \delta\left(t - t_{sp}\right)$$

$$\frac{d\mathrm{g}_E}{dt} = -\frac{\mathrm{g}_E}{\tau_E} + \bar{\mathrm{g}}_E u^+ x^- \delta\left(t - t_{sp}\right)$$

- U_0: constant, synaptic release probability at rest
- τ_f: synaptic facilitation time constant of u
- τ_d: synaptic depression time constant of x
- τ_E: synaptic time constant of g_E
- $\bar{\mathrm{g}}_E$: synaptic conductance strength
- + and – superscripts denote the time just after or just before the presynaptic spike

Phenomenological model



Adopted from http://www.scholarpedia.org/article/Short-term_synaptic_plasticity

- Following a presynaptic spike
  - (i) the fraction u (release probability) of the available pool to be utilized increases due to spike-induced calcium influx to the presynaptic terminal, after which
  - (ii) u is consumed to increase the post-synaptic conductance. Between spikes, u decays back to zero with time constant τ_f and x recovers to 1 with time constant τ_d.

Next week

- HH models and how they relate to previously seen models

- Cable theory